

TASK 11 – Recommend and Defend the Final Design

Based on your mathematical analysis and experimental findings, recommend the most appropriate algorithmic solution for the university student-record system.

Discuss:

- Which approach you recommend
 - Why it is more appropriate than the alternatives
 - Its theoretical efficiency
 - Its experimentally observed performance
 - Any preprocessing cost involved
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- Relevant trade-offs or limitations
 - Whether the solution remains practical as the database grows

Your recommendation must be based on evidence obtained from your own analysis

and experimentation.

Simply identifying the algorithm with the lowest Big-O complexity is not sufficient.

Evidence Expected

A technically justified final design recommendation supported by both mathematical

analysis and experimental evidence.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement

The university now needs to process 50,000 search requests per hour on a database

containing 1,000,000 student records.

However, new student records are added only once per day.

Re-evaluate your original design in response to these changed operating conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Discuss:

- What has changed from the original problem
- Which original assumptions are affected
- Why the frequency of searches is now important
- Why the relatively low frequency of record updates may influence the design

Evidence Expected

A clear analysis of how the changed requirements affect the original algorithmic design.

B. Evaluation of the Original Design – 2 Marks

Evaluate whether your original recommended solution remains appropriate.

Discuss:

- Its expected behaviour under 50,000 searches per hour

- Whether its preprocessing cost is acceptable under the new conditions
- Whether another approach could provide better overall performance

Evidence Expected

A technically supported evaluation of the suitability of the original design under the new operating conditions.

C. Adaptation or Redesign – 2 Marks

Decide whether to:

Retain, modify or redesign the original solution.

If you modify or redesign it:

- Describe the revised algorithmic strategy
- Explain what has changed
- Determine the effect on computational complexity

If you retain the original solution, provide sufficient technical justification for why no redesign is necessary.

Evidence Expected

An appropriate design response to the changed requirement, with clear algorithmic reasoning.

D. Evidence-Based Defence – 2 Marks

Defend your final design decision using evidence.

Use:

- Mathematical complexity analysis
- Experimental results, where appropriate

- Search frequency
- Preprocessing cost
- Dataset size
- Relevant computational trade-offs

Your final decision must follow logically from your analysis.

Evidence Expected

A well-supported final design decision integrating theoretical analysis, experimental

evidence and the changed system constraints.